

XIAO HE

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ACADEMIC APPOINTMENTS

Tenure-Track Associate Professor <i>School of Aeronautics and Astronautics, Shanghai Jiao Tong University</i>	02/2025 – present <i>Shanghai, China</i>
Postdoctoral Research Assistant/Associate <i>UQ Lab, Department of Aeronautics, Imperial College London</i> <i>Advisor: Prof. Francesco Montomoli, Dr. Vittorio Michelassi</i>	08/2022 – 01/2025 <i>London, UK</i>

EDUCATION

Imperial College London <i>Ph.D. in Mechanical Engineering</i> <i>Advisor: Prof. Mehdi Vahdati, Prof. Ricardo Martinez-Botas</i>	09/2018 – 03/2023 <i>London, UK</i>
Institute of Science Tokyo <i>Exchange research student in Mechanical Engineering</i> <i>Advisor: Prof. Hiroto Tanaka</i>	07/2017 – 09/2017 <i>Tokyo, Japan</i>
Tsinghua University <i>M.S. (Hons) in Power Engineering and Engineering Thermophysics</i> <i>Advisor: Prof. Xinqian Zheng</i>	09/2015 – 06/2018 <i>Beijing, China</i>
Tsinghua University <i>B.E. (Hons) in Vehicle Engineering</i>	08/2011 – 07/2015 <i>Beijing, China</i>

AWARDS AND HONORS

GPPS Early Career Award	2024
Imperial College Katopodis Prize (best PhD thesis within the Thermofluids division)	2023
ASME IGTI Turbomachinery Committee Best Paper Finalist	2022
Chinese Government Award for Outstanding Self-Financed Students Abroad	2022
Great Britain-China Educational Trust Grant	2021
Henry Lester Trust Grant	2021
ASME IGTI Turbo Expo Early Career Engineer Travel Award	2020,2021,2024
ASME IGTI Student Advisory Committee Travel Award	2019
Imperial College President's PhD Scholarship	2018
China National Scholarship	2017
Japan Student Services Organization Scholarship	2017
Overall Excellent Scholarship	2016
Tsinghua University Honored Graduate Award	2015
Tsinghua University Best Undergraduate Thesis Award	2015
Academic Excellent Scholarship	2012,2013,2014
1 st Prize in the 29 th National College Student Physics Competition (Beijing)	2012
1 st Prize in the 27 th National High School Student Physics Competition (Hunan)	2010

FUNDED PROJECTS

NSFC Young Scientist Fund	2026-2028
NSFC Excellent Young Scientists Fund Program (Overseas)	2026-2028

TEACHING EXPERIENCE

Lecturer , Jet Propulsion Principle (undergraduate Year 3) SJTU	03/2026 – now
Supervisor , undergraduate and graduate students SJTU	09/2025 – now
Graduate Teaching Assistant , Fluid Mechanics (undergraduate Year 2) Imperial	10/2019 – 06/2021
Mentor , Master Thesis and Research Internship Imperial	01/2020 – 09/2023
Mentor , Undergraduate Thesis Tsinghua	01/2016 – 06/2018

INVITED TALKS

1. Invited conference talk: "Progress on Data-Driven Turbulence/Transition Modeling for Accurate Performance Prediction of Aero-Engine Components"
 - The 10th China Joint Conference of Aerospace Propulsion (CJCAP), Huzhou, China. (07/2026)
 - The 3rd Aerospace Frontiers Conference (AFC 2026), Nanjing, China (04/2026).
2. Seminar talk: "Progress of Data-Driven Turbulence Modeling for Compressor Stalled/Separated Flows"
 - School of Power and Energy, Northwestern Polytechnical University (05/2025).
 - School of Energy and Power Engineering, Xi'an Jiao Tong University (05/2025).
 - School of Energy and Power Engineering, Beihang University (04/2025).
 - Digital Twin Research Center, Institute of Engineering Thermophysics, Chinese Academy of Science (04/2025).
3. Seminar talk: "Data-Driven RANS Turbulence Modeling for Compressor Stall"
 - School of Aeronautics and Astronautics, Shanghai Jiao Tong University (06/2023).
 - Institute of Gas Turbines and Aerospace Propulsion, Technische Universität Darmstadt (06/2022).
 - Institute of Turbomachinery, School of Mechanical Engineering, Shanghai Jiao Tong University (04/2022).

PUBLICATIONS

I have authored over 30 peer-reviewed research papers including 23 journal papers as the first/correspondence author. My [Google Scholar](#) statistics are: Citation $\geq$ 800, h-index $\geq$ 17; my [WOS](#) statistics are: Citation $\geq$ 602, h-index $\geq$ 15. A full list of publication is provided below. (* corresponding author; † equal contribution)

Peer-Reviewed Journal Publications

1. **He, X.***, Zhao, F., and Vahdati, M., "Spatio-Temporal Evolution of Tip Leakage Flow in a Low-Speed Axial Compressor," **ASME Journal of Turbomachinery** (accepted).
2. **He, X.***, Li, B., Wang, J., Fu, Y., and Teng, J., "Data-Driven Recalibration of Shear Stress Transport Intermittency Model for Transitional Compressor," **Chinese Journal of Aeronautics** (accepted).
3. Song, W., Wang, Z., Tao, Y., **He, X.***, and Kou, J.*. "Research progress and outlook on intelligent fluid mechanics for aeroengines," **Journal of Propulsion Technology**, 2026, 47(6), 25-52. (in Chinese)
4. Chen, C., Zhu, M.*, **He, X.**, Shi, H., Deng, H., Teng, J., and Qiang, X., "Loss Characteristics of Cantilevered Cascade With Moving Endwall Under Upstream Wake Influence," **ASME Journal of Turbomachinery**, 2026, 148(7), 071012. [\[doi\]](#)
5. **He, X.***, Montomoli, F., Michelassi, V., Panizza, A., and Pulga, L., "Multiscale Data-Driven RANS Closure for Trailing Edge Cutback Film Cooling Flows," **ASME Journal of Turbomachinery**, 2026, 148(5), 051003. [\[doi\]](#)
6. Zhang, G., Zhu, M.*, He, Z., Yuan, Y., **He, X.**, Jiang, H., and Teng, J., "PIV Experiment Research of Leakage Flow in Open Clearance Compressor Cascade," **Journal of Engineering Thermophysics**, 2025, 46(12), 3961-3969. (in Chinese) [\[url\]](#)
7. Shao, R., **He, X.***, Zhu, M., Klausmann, F., and Teng, J., "Loss Mechanisms of Shrouded Stator Cavity Flow in an Axial Compressor," **AIAA Journal of Propulsion and Power**, 2025, 41(4), 503-516. [\[doi\]](#)
8. Shao, R., **He, X.**, Zhu, M.*, Teng, J., Ju, Z., and Fan, L., "Mechanisms of Cavity Windage Heating on the Performance of an Axial Compressor," **Physics of Fluids**, 2025, 37, 036125. [\[doi\]](#)
9. **He, X.***, and Klausmann, F., "RANS Capabilities for Transonic Axial Compressor: A Perspective from GPPS CFD Workshop," **ASME Journal of Turbomachinery**, 2025, 147(2), 021007. [\[preprint\]](#) [\[doi\]](#)
10. **He, X.***, Montomoli, F., Michelassi, V., Panizza, A., and Pulga, L., "Characteristics of Deterministic and Stochastic Unsteadiness of Trailing Edge Cutback Film Cooling Flows," **ASME Journal of Turbomachinery**, 2025, 147(1), 011009. [\[doi\]](#)
11. Klausmann, F.*†, Kilian, N.†, **He, X.**, Franke, D., Schmidt, B., and Schiffer, H., "Transonic Compressor Darmstadt Open Test Case: Experimental Investigation of Stator Secondary Flows and Hub Leakage," **ASME Journal of Turbomachinery**, 2024, 146(10), 101007. [\[doi\]](#)

12. Yan, C., Wang, B., **He, X.**, Zhao, F., Zheng, X., Vahdati, M., and Zheng, X.*, “Extension and Validation of the Turbomachinery Capabilities of SU2 Open Source CFD Code,” **ASME Journal of Turbomachinery**, 2024, 146(6), 061003. [\[doi\]](#)
13. Shao, R., **He, X.***, Zhu, M., Klausmann, F., and Teng, J., “Characterizing Shrouded Stator Cavity Flow on the Performance of a Single-Stage Axial Transonic Compressor,” **ASME Journal of Turbomachinery**, 2023, 145(11), 111004. [\[preprint\]](#) [\[doi\]](#)
14. Qiao, B., **He, X.***, Vahdati, M., Ju, Y., and Zhang, C., “On the Over-Prediction of Centrifugal Compressor Pressure Ratio in the High Impeller Tip Mach Number Regime,” **ASME Journal of Turbomachinery**, 2023, 145(10), 104502. [\[preprint\]](#) [\[doi\]](#)
15. Xia, K., **He, X.***, Zhu, M., Klausmann, F., Teng, J., and Vahdati, M., “Endwall Geometric Uncertainty and Error on the Performance of TUDA-GLR-OpenStage Transonic Axial Compressor,” **Journal of the Global Power and Propulsion Society**, 2023, 7, 113-126. [\[preprint\]](#) [\[doi\]](#)
16. **He, X.***†, Zhu, M.†, Xia, K., Klausmann, F., Teng, J., and Vahdati, M., “Validation and Verification of RANS Solvers for TUDA-GLR-OpenStage Transonic Axial Compressor,” **Journal of the Global Power and Propulsion Society**, 2023, 7, 13-29. [\[preprint\]](#) [\[doi\]](#)
17. **He, X.***, Zhao, F., and Vahdati, M., “A Turbo-Oriented Data-Driven Modification to the Spalart-Allmaras Turbulence Model,” **ASME Journal of Turbomachinery**, 2022, 144(12), 121007. [\[preprint\]](#) [\[doi\]](#)
18. Wang, R.†, **He, X.**†, and Yan, X., “Spectral Proper Orthogonal Decomposition of Trailing Edge Cutback Film Cooling Flows,” **Physics of Fluids**, 2022, 34(10), 105106. [\[preprint\]](#) [\[doi\]](#)
19. **He, X.***, Tan, J., Rigas, G., and Vahdati, M., “On the Explainability of Machine-Learning-Assisted Turbulence Modeling for Transonic Flows,” **International Journal of Heat and Fluid Flow**, 2022, 97, 109038. [\[preprint\]](#) [\[doi\]](#)
20. Wang, J., **He, X.**, Wang, B., and Zheng, X., “Shapley Additive Explanations of Multi-Geometrical Variable Coupling Effect in Transonic Compressor,” **ASME Journal of Engineering for Gas Turbines and Power**, 2022, 144(4), 041015. [\[preprint\]](#) [\[doi\]](#)
21. **He, X.***, Zhao, F., and Vahdati, M., “Detached Eddy Simulation: Recent Development and Application to Compressor Tip Leakage Flow,” **ASME Journal of Turbomachinery**, 2022, 144(1), 011009. [\[preprint\]](#) [\[doi\]](#)
22. **He, X.***, Fang, Z., Rigas, G., and Vahdati, M., “Spectral Proper Orthogonal Decomposition of Compressor Tip Leakage Flow,” **Physics of Fluids**, 2021, 33(10), 105105. [\[preprint\]](#) [\[doi\]](#)
23. **He, X.***, Zhao, F., and Vahdati, M., “Uncertainty Quantification of Spalart-Allmaras Turbulence Model Coefficients for Compressor Stall,” **ASME Journal of Turbomachinery**, 2021, 143(8), 081007. [\[preprint\]](#) [\[doi\]](#)
24. **He, X.***, Zhao, F., and Vahdati, M., “Uncertainty Quantification of Spalart-Allmaras Turbulence Model Coefficients for Simplified Compressor Flow Features,” **ASME Journal of Fluids Engineering**, 2020, 142(9), 091501. [\[preprint\]](#) [\[doi\]](#)
25. Zhang, W., **He, X.**, Wang, B., Sun, Z., and Zheng, X., “Stability Improvement of a High Pressure Ratio Centrifugal Compressor by Flow Injection,” **ASCE Journal of Aerospace Engineering**, 2020, 33(6), 04020072. [\[doi\]](#)
26. Zou, W., **He, X.**, Zhang, W., Niu, Z., and Zheng, X., “Roles of Vanes in Diffuser on Stability of Centrifugal Compressor,” **Proceedings of the Institution of Mechanical Engineers, Part G: Journal of Aerospace Engineering**, 2019, 233(14), 5380-5392. [\[doi\]](#)
27. **He, X.**, and Zheng, X., “Roles and Mechanisms of Casing Treatment on Different Scales of Flow Instability in High Pressure Ratio Centrifugal Compressors,” **Aerospace Science and Technology**, 2019, 84, 734-746. [\[preprint\]](#) [\[doi\]](#)
28. **He, X.**, and Zheng, X., “Flow Instability Evolution in High Pressure Ratio Centrifugal Compressor with Vaned Diffuser,” **Experimental Thermal and Fluid Science**, 2018, 98, 719-730. [\[preprint\]](#) [\[doi\]](#)
29. **He, X.**, and Zheng, X., “Performance Improvement of Transonic Centrifugal Compressors by Optimization of Complex Three-Dimensional Features,” **Proceedings of the Institution of Mechanical Engineers, Part G: Journal of Aerospace Engineering**, 2017, 231(14), 2723-2738. [\[preprint\]](#) [\[doi\]](#)
30. **He, X.**, and Zheng, X., “Mechanisms of Sweep on the Performance of Transonic Centrifugal Compressor Impellers,” **Applied Sciences**, 2017, 7(10), 1081. [\[doi\]](#)
31. Hosseini, M., Sun, Z., **He, X.**, and Zheng, X., “Effects of Radial Gap Ratio between Impeller and Vaned Diffuser on Performance of Centrifugal Compressors,” **Applied Sciences**, 2017, 7(7): 728. [\[doi\]](#)
32. Ebrahimi, M., Huang, Q., **He, X.**, and Zheng, X., “Effects of Variable Diffuser Vanes on Performance of a Centrifugal Compressor with Pressure Ratio of 8.0,” **Energies**, 2017, 10(5), 682. [\[doi\]](#)
33. **He, X.**, and Zheng, X., “Mechanisms of Lean on the Performance of Transonic Centrifugal Compressor Impellers,” **AIAA Journal of Propulsion and Power**, 2016, 32(5), 1220-1229. [\[preprint\]](#) [\[doi\]](#)

Peer-Reviewed Conference Publications

1. **He, X.***, Zhao, F., and Vahdati, M., “Spatio-Temporal Evolution of Tip Leakage Flow in a Low-Speed Axial Compressor,” Proceedings of the ASME Turbo Expo 2026: Turbomachinery Technical Conference and Exposition. ASME Paper No. GT2026-176359. (accepted)
2. Li, B., Wang, J., Fu, Y., Teng, J., and **He, X.***, “Investigation of Performance Evolution Mechanism of Low-Reynolds Number Compressors,” 2025 Annual Academic Conference of the Chinese Society of Engineering Thermophysics, No. 251685. (in Chinese)
3. **He, X.***, Montomoli, F., Michelassi, V., Panizza, A., and Pulga, L., “Multiscale Data-Driven RANS Closure for Trailing Edge Cutback Film Cooling Flows,” Proceedings of the ASME Turbo Expo 2025: Turbomachinery Technical Conference and Exposition. ASME Paper No. GT2025-151761. [\[doi\]](#)
4. Seki, R., Terauchi, Y., Zhao, F., **He, X.**, and Vahdati, M., “Optimisation of SA-PG ω Turbulence Model for Multi-Stage Compressor at Off-Design Conditions,” Proceedings of the ASME Turbo Expo 2025: Turbomachinery Technical Conference and Exposition. ASME Paper No. GT2025-152022. [\[doi\]](#)
5. Chen, C., Zhu, M., **He, X.**, Shi, H., Deng, H., Teng, J., and Qiang, X., “Study on the Loss Characteristics and Flow Mechanism of Cantilevered Cascade With Moving Endwall Under the Influence of Upstream Wake,” Proceedings of the ASME Turbo Expo 2025: Turbomachinery Technical Conference and Exposition. ASME Paper No. GT2025-152941. [\[doi\]](#)
6. **He, X.***, Montomoli, F., Michelassi, V., Panizza, A., and Pulga, L., “Characteristics of Deterministic and Stochastic Unsteadiness of Trailing Edge Cutback Film Cooling Flows,” Proceedings of the ASME Turbo Expo 2024: Turbomachinery Technical Conference and Exposition. Volume 7: Heat Transfer: Combustors; Heat Transfer: Film Cooling. London, United Kingdom. June 24–28, 2024. V007T12A011. ASME Paper No. GT2024-123002. [\[doi\]](#)
7. Deng, H., **He, X.***, Zhu, M., Klausmann, F., and Teng, J., “Unsteady Flow Phenomenon in the TUDa-GLR-OpenStage Compressor: URANS Observations,” Proceedings of the Global Power and Propulsion Society Hong Kong Conference 2023. Hong Kong, China. October 17-19, 2023. GPPS Paper No. GPPS-TC-2023-0027. [\[doi\]](#)
8. Shao, R., **He, X.***, Zhu, M., Klausmann, F., and Teng, J., “Characterizing Shrouded Stator Cavity Flow on the Performance of a Single-Stage Axial Transonic Compressor,” Proceedings of the ASME Turbo Expo 2023: Turbomachinery Technical Conference and Exposition. Volume 2D: Turbomachinery. Boston, Massachusetts, USA. June 26–30, 2023. ASME Paper No. GT2023-101131. [\[preprint\]](#) [\[doi\]](#)
9. Yan, C., Wang, B., **He, X.**, Zhao, F., Zheng, X., Vahdati, M.*, and Zheng, X.*, “Extension and Validation of the Turbomachinery Capabilities of SU2 Open Source CFD Code,” Proceedings of the ASME Turbo Expo 2023: Turbomachinery Technical Conference and Exposition. Volume 2D: Turbomachinery. Boston, Massachusetts, USA. June 26–30, 2023. ASME Paper No. GT2023-103544. [\[doi\]](#)
10. Klausmann, F.*†, Kilian, N.†, **He, X.**, Franke, D., Schmidt, B., and Schiffer, H., “Transonic Compressor Darmstadt Open Test Case: Experimental Investigation of Stator Secondary Flows and Hub Leakage,” Proceedings of the ASME Turbo Expo 2023: Turbomachinery Technical Conference and Exposition. Volume 2D: Turbomachinery. Boston, Massachusetts, USA. June 26–30, 2023. ASME Paper No. GT2023-103502. [\[doi\]](#)
11. Xia, K., **He, X.***, Zhu, M., Klausmann, F., Teng, J., and Vahdati, M., “Endwall Geometric Uncertainty and Error on the Performance of TUDA-GLR-OpenStage Transonic Axial Compressor,” Proceedings of the Global Power and Propulsion Society Chania Conference 2022. Chania, Greece. September 12-14, 2022. GPPS Paper No. GPPS-TC-2022-0126. [\[preprint\]](#) [\[doi\]](#)
12. **He, X.***, Zhao, F., and Vahdati, M., “A Turbo-Oriented Data-Driven Modification to the Spalart-Allmaras Turbulence Model,” Proceedings of the ASME Turbo Expo 2022: Turbomachinery Technical Conference and Exposition. Volume 2D: Turbomachinery. Rotterdam, Netherlands. June 13–17, 2022. ASME Paper No. GT2022-80456. **ASME IGTI Turbomachinery Committee Best Paper Finalist.** [\[preprint\]](#) [\[doi\]](#)
13. Zhu, M., **He, X.***, Klausmann, F., Teng, J., and Vahdati, M., “Validation and Verification of RANS Solvers for TUDA-GLR-OpenStage Transonic Axial Compressor,” Proceedings of the Global Power and Propulsion Society Xi’an Conference 2021. Virtual, Online. April 11-13, 2022. GPPS Paper No. GPPS-TC-2021-0050. [\[preprint\]](#) [\[doi\]](#)
14. **He, X.*†**, Fang, Z.†, Vahdati, M., and Rigas, G., “Spectral Proper Orthogonal Decomposition of Compressor Tip Leakage Flow,” Proceedings of the 14th European Conference on Turbomachinery Fluid Dynamics & Thermodynamics. Virtual, Online. April 12-14, 2021. ETC Paper No. ETC2021-491. [\[preprint\]](#) [\[doi\]](#)
15. Tan, J.†, **He, X.*†**, Vahdati, M., and Rigas, G., “Towards Explainable Machine Learning Assisted Turbulence Modelling for Transonic Flows,” Proceedings of the 14th European Conference on Turbomachinery Fluid Dynamics & Thermodynamics. Virtual, Online. April 12-14, 2021. ETC Paper No. ETC2021-490. [\[preprint\]](#) [\[doi\]](#)
16. **He, X.***, Zhao, F., and Vahdati, M., “Detached Eddy Simulation: Recent Development and Application to Compressor Tip Leakage Flow,” Proceedings of the ASME Turbo Expo 2021: Turbomachinery Technical Conference and Exposition. Volume 2C: Turbomachinery. Virtual, Online. June 7–11, 2021. V02CT34A011. ASME Paper No. GT2021-59077. [\[preprint\]](#) [\[doi\]](#)

17. **He, X.***, Zhao, F., and Vahdati, M., “Evaluation of Spalart-Allmaras Turbulence Model Forms for a Transonic Axial Compressor,” Proceedings of the Global Power and Propulsion Society Chania Conference 2020. Virtual, Online. September 7-9, 2020. GPPS Paper No. GPPS-CH-2020-0013. [\[preprint\]](#) [\[doi\]](#)
18. **He, X.***, Zhao, F., and Vahdati, M., “Uncertainty Quantification of Spalart-Allmaras Turbulence Model Coefficients for Compressor Stall,” Proceedings of the ASME Turbo Expo 2020: Turbomachinery Technical Conference and Exposition. Volume 2C: Turbomachinery. Virtual, Online. September 21–25, 2020. V02CT35A028. ASME Paper No. GT2020-15014. [\[preprint\]](#) [\[doi\]](#)
19. **He, X.***, Zhao, F., and Vahdati, M., “Machine Learning Uncertainty Quantification of Spalart-Allmaras Turbulence Model for Compressors,” Proceedings of the Global Power and Propulsion Society Beijing Conference 2019. Beijing, China. September 16-18, 2019. GPPS Paper No. 2019-BJ-0050. [\[preprint\]](#) [\[doi\]](#)
20. **He, X.**, Zheng, X., Wei, J., and Zeng, H., “Investigation of Vaned Diffuser Splitters on the Performance and Flow Control of High Pressure Ratio Centrifugal Compressors,” Proceedings of the ASME Turbo Expo 2016: Turbomachinery Technical Conference and Exposition. Volume 2D: Turbomachinery. Seoul, South Korea. June 13–17, 2016. V02DT42A008. ASME Paper No. GT2016-56255. [\[doi\]](#)

SOFTWARES

HADES: an unstructured finite volume CFD solver for internal and external flows.

PyMeshHADES: a Pythonic mesh generation and conversion tool for the in-house solver HADES.

SPOD Python: a Pythonic realization of SPOD and its applications to some fluid mechanics problems.

TurbAna: a Python toolkit that calculates turbulence anisotropy and turbulent viscosity from Reynolds stress.

ACADEMIC SERVICE

Editorial Service

Youth Editorial Member, Theor Appl Mech Lett (2025-2030) Guest Editor, Energies (2025)

Conference Session Chair / Track Chair

ASME Turbo Expo: Heat Transfer (2024-2025)

GPPS Conf.: CFD Workshop (2021-2025)

GPPS Conf.: Turbomachinery Aerodynamics (2024-2026)

GPPS Conf.: Turbomachinery Aeroelasticity (2024)

Reviewer for Journals and Conferences

I have reviewed 107 manuscripts from 38 journals and 2 conferences, and more details can be found in my [WOS](#) account. The full list of reviewed journals and conferences is as follows (in alphabetic order).

Acta Astronaut	Chinese J Aeronaut	GPPS Conf	J Eng Thermophys
Adv Mech Eng	Comm Nonlin Sci Num Sim	IMechE A: J Power Energy	J Glob Power Propuls Soc
Aerosp Sci Technol	Comput Fluids	IMechE C: J Mechan Eng Sci	J Mar Sci Eng
Aerospace	Discov Appl Sci	IMechE D: J Automob	Math Comput Simul
AIAA J	Energies	IMechE G: J Aerosp Eng	Measurement
AIAA J Aircr	Energy	Int J Aeronaut Space Sci	Ocean Eng
ASCE J Aerosp Eng	Energy Convers Manag	Int J Heat Fluid Flow	Phys Fluids
ASME J Fluids Eng	Energy Sci Eng	Int J Mech Sci	Renew Energy
ASME J Turbomach	Entropy	Int J Therm Sci	Results Eng
ASME Turbo Expo	Fluids	J Appl Fluid Mech	Sci Rep